

School of Natural Sciences and Mathematics

The Biology Program at UT Dallas emphasizes the unifying molecular and cellular nature of organisms. At the center of the Biology undergraduate curriculum are the biochemical, genetic, and cell biology concepts and tools used to study the genes of prokaryotes and eukaryotes, to study the proteins and ribonucleic acids (RNA) encoded by these genes, and to study how the expression of these genes is regulated during the development and lifetimes of organisms. Molecular Biology represents a fusion of the four disciplines of biochemistry, biophysics, genetics, and cell biology. Modern biology requires a background in other disciplines such as chemistry, mathematics, physics, and computer sciences. Principles from these disciplines have to be merged to understand and apply new biotechnology and genetic engineering techniques. It is desirable for entering students to have a broad interest and background in the sciences.

Biology (BA, BS)

Both BS and BA degrees are offered in Biology at UT Dallas. The BS degrees are intended as preparation for scientific careers in biology or careers in the health professions. The BA degree is intended as a liberal arts biology major with less emphasis on calculus and more semester credit hours free for coursework in other disciplines. Biology offers a streamlined double major with Business Administration or Criminology. [Accelerated BS/MS Fast Track programs](#) are available that provide the opportunity to partially satisfy the requirements of a master's degree while completing the bachelor's degree in Biology.

Bachelor of Arts in Biology

[Degree Requirements](#) (120 semester credit hours)

[View an Example of Degree Requirements by Semester](#)

Faculty

FACG> nsm-biology-ba

Professors: Juan E. González, Kelli Palmer, Lawrence J. Reitzer, Donal Skinner, Stephen Spiro, Li Zhang, Michael Qiwei Zhang

Associate Professors: Joseph Boll, Nicole De Nisco, Nikki Delk, Tian Hong, Tae Hoon Kim, Faruck Morcos, Duane D. Winkler, Zhenyu Xuan

Assistant Professors: Nicholas Dillon, Xintong Dong, Lin Jia, Purna Joshi, Brandon Kim, Erica Sanchez, Darshan Sapkota, Yuki Shindo, jxs230052

Professors Emeriti: Lee A. Bulla, Donald M. Gray

Associate Professors Emeriti: Gail A. M. Breen, Dennis L. Miller

Clinical Professor: David Murchison, Eberhard Voit

Professors of Instruction: Scott Rippel, Uma Srikanth

Associate Professors of Instruction: Mehmet Candas, Meenakshi Maitra, Jing Pan, Elizabeth Pickett, Michelle Wilson, Wen-Ho Yu

Assistant Professors of Instruction: Stephanie Boyd, Anne Davenport, Amy Jo Gomez, Yi Huang, Li Liu, Caitlin Maynard, Simbarashe Mazambani, Ramesh Padmanabhan, Jing Pan, Ruben Ramirez, Eva Sadat, Subha Sarcar, Michelle Wilson, Zhuoru Wu

Research Assistant Professor: Ru-Hung Wang

Lecturer: Kathleen McRoy

I. Core Curriculum Requirements: 42 semester credit hours¹

Communication: 6 semester credit hours

Select any 6 semester credit hours from [Communication Core](#) courses (see advisor)

Mathematics: 3 semester credit hours

Choose one course from the following:

[MATH 2413](#) Differential Calculus^{2, 3, 4}

or [MATH 2417](#) Calculus I^{2, 3, 4}

[MATH 1325](#) Applied Calculus I² (BA only)

Or select any 3 semester credit hours from [Mathematics Core](#) courses (see advisor)

Life and Physical Sciences: 6 semester credit hours

[CHEM 1311](#) General Chemistry I²

or [CHEM 1315](#) Honors Freshman Chemistry I²

[CHEM 1312](#) General Chemistry II²

or [CHEM 1316](#) Honors Freshman Chemistry II²

Or select any 6 semester credit hours from [Life and Physical Sciences Core](#) courses (see advisor)

Language, Philosophy and Culture: 3 semester credit hours

Select any 3 semester credit hours from [Language, Philosophy and Culture Core](#) courses (see advisor)

Creative Arts: 3 semester credit hours

Select any 3 semester credit hours [Creative Arts Core](#) courses (see advisor)

American History: 6 semester credit hours

Select any 6 semester credit hours from [American History Core](#) courses (see advisor)

Government / Political Science: 6 semester credit hours

Select any 6 semester credit hours from [Government/Political Science Core](#) courses (see advisor)

Social and Behavioral Sciences: 3 semester credit hours

Select any 3 semester credit hours from [Social and Behavioral Sciences Core](#) courses (see advisor)

Component Area Option: 6 semester credit hours

Choose two courses from the following:

[BIOL 2311](#) Introduction to Modern Biology I ^{2, 5} _{_}

[MATH 2414](#) Integral Calculus ^{2, 3, 4} _{_ _}

or [MATH 2419](#) Calculus II ^{2, 3, 4} _{_ _}

or [STAT 2332](#) Introductory Statistics for Life Sciences ² _{_}

Or select any 6 semester credit hours from [Component Area Option Core](#) courses (see advisor)

II. Major Requirements: 53-56 semester credit hours beyond Core Curriculum

Major Preparatory Courses: 18-21 semester credit hours beyond Core Curriculum

[CHEM 1311](#) General Chemistry I ² _{_}

or [CHEM 1315](#) Honors Freshman Chemistry I ² _{_}

[CHEM 1111](#) General Chemistry Laboratory I

or [CHEM 1115](#) Honors Freshman Chemistry Laboratory I

[CHEM 1312](#) General Chemistry II₂

or [CHEM 1316](#) Honors Freshman Chemistry II₂

[CHEM 1112](#) General Chemistry Laboratory II

or [CHEM 1116](#) Honors Freshman Chemistry Laboratory II

[CHEM 2323](#) Introductory Organic Chemistry I₅

or [CHEM 2327](#) Honors Organic Chemistry I₅

[CHEM 2325](#) Introductory Organic Chemistry II₅

or [CHEM 2328](#) Honors Organic Chemistry II₅

[CHEM 2233](#) Introductory Organic Chemistry Laboratory₅

or [CHEM 2237](#) Honors Organic Chemistry Laboratory₅

[MATH 2413](#) Differential Calculus_{1, 2, 3, 4}

and [MATH 2414](#) Integral Calculus_{1, 2, 3, 4}

or [MATH 2417](#) Calculus I_{1, 2, 3, 4}

and [MATH 2419](#) Calculus II_{1, 2, 3, 4}

or [MATH 1325](#) Applied Calculus I_{1, 2, 3, 4}

and [STAT 2332](#) Introductory Statistics for Life Sciences_{2, 3}

[PHYS 2325](#) Mechanics_{1, 2}

and [PHYS 2125](#) Physics Laboratory I_{1, 2}

or [PHYS 2421](#) Honors Physics I - Mechanics and Heat₆

or [PHYS 1301](#) College Physics I

and [PHYS 1101](#) College Physics Laboratory I

[PHYS 2326](#) Electromagnetism and Waves

and [PHYS 2126](#) Physics Laboratory II

or [PHYS 2422](#) Honors Physics II - Electromagnetism and Waves

and [PHYS 2126](#) Physics Laboratory II

or [PHYS 1302](#) College Physics II

and [PHYS 1102](#) College Physics Laboratory II

Major Core Courses: 26-29 semester credit hours beyond Core Curriculum

[BIOL 2281](#) Introductory Biology Laboratory⁵

[BIOL 2111](#) Introduction to Modern Biology Workshop I⁵

[BIOL 2112](#) Introduction to Modern Biology Workshop II⁵

[BIOL 2311](#) Introduction to Modern Biology I^{2, 5}

[BIOL 2312](#) Introduction to Modern Biology II⁵

[BIOL 3401](#) Genetics

[BIOL 3402](#) Molecular and Cell Biology

[BIOL 3461](#) Biochemistry I

or [CHEM 3461](#) Biochemistry I

[BIOL 3462](#) Biochemistry II

or [CHEM 3462](#) Biochemistry II

or [BIOL 3335](#) Microbial Physiology

[BIOL 3380](#) Biochemistry Laboratory

Major Related Courses: 9 semester credit hours⁷

9 semester credit hours of upper-division [BIOL](#) electives

III. Elective Requirements: 22-25 semester credit hours

The plan must include sufficient upper-division courses to total 45 upper-division semester credit

hours.

Bachelor of Science in Biology

[Degree Requirements](#) (120 semester credit hours)

[View an Example of Degree Requirements by Semester](#)

Faculty

FACG> nsm-biology-bs

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Research Assistant Professor: Ru-Hung Wang

Lecturer: Kathleen McRoy

I. Core Curriculum Requirements: 42 semester credit hours^{[1](#)}

Communication: 6 semester credit hours

Select any 6 semester credit hours from [Communication Core](#) courses (see advisor)

Mathematics: 3 semester credit hours

Choose one course from the following:

[MATH 2413](#) Differential Calculus ^{2, 3, 4} _{_ _}

or [MATH 2417](#) Calculus I ^{2, 3, 4} _{_ _}

[MATH 1325](#) Applied Calculus I ² _{_} (BA only)

Or select any 3 semester credit hours from [Mathematics Core](#) courses (see advisor)

Life and Physical Sciences: 6 semester credit hours

[CHEM 1311](#) General Chemistry I ² _{_}

or [CHEM 1315](#) Honors Freshman Chemistry I ² _{_}

[CHEM 1312](#) General Chemistry II ² _{_}

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Or select any 6 semester credit hours from [Life and Physical Sciences Core](#) courses (see advisor)

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Select any 3 semester credit hours from [Language, Philosophy and Culture Core](#) courses (see advisor)

Creative Arts: 3 semester credit hours

Select any 3 semester credit hours [Creative Arts Core](#) courses (see advisor)

American History: 6 semester credit hours

Select any 6 semester credit hours from [American History Core](#) courses (see advisor)

Government / Political Science: 6 semester credit hours

Select any 6 semester credit hours from [Government/Political Science Core](#) courses (see advisor)

Social and Behavioral Sciences: 3 semester credit hours

Select any 3 semester credit hours from [Social and Behavioral Sciences Core](#) courses (see advisor)

Component Area Option: 6 semester credit hours

Choose two courses from the following:

[BIOL 2311](#) Introduction to Modern Biology I 2, 5

[MATH 2414](#) Integral Calculus 2, 3, 4

or [MATH 2419](#) Calculus II 2, 3, 4

Or select any 6 semester credit hours from [Component Area Option Core](#) courses (see advisor)

II. Major Requirements: 59-62 semester credit hours beyond Core Curriculum

Major Preparatory Courses: 18-21 semester credit hours beyond Core Curriculum

[CHEM 1311](#) General Chemistry I 2

or [CHEM 1315](#) Honors Freshman Chemistry I 2

[CHEM 1111](#) General Chemistry Laboratory I

or [CHEM 1115](#) Honors Freshman Chemistry Laboratory I

[CHEM 1312](#) General Chemistry II 2

or [CHEM 1316](#) Honors Freshman Chemistry II 2

[CHEM 1112](#) General Chemistry Laboratory II

or [CHEM 1116](#) Honors Freshman Chemistry Laboratory II

[CHEM 2323](#) Introductory Organic Chemistry I 5

or [CHEM 2327](#) Honors Organic Chemistry I 5

[CHEM 2325](#) Introductory Organic Chemistry II 5

or [CHEM 2328](#) Honors Organic Chemistry II 5

[CHEM 2233](#) Introductory Organic Chemistry Laboratory 5

or [CHEM 2237](#) Honors Organic Chemistry Laboratory 5

[MATH 2413](#) Differential Calculus 1, 2, 3, 4

and [MATH 2414](#) Integral Calculus 1, 2, 3, 4

or [MATH 2417](#) Calculus I ^{1, 2, 3, 4} _{_ _ _}

and [MATH 2419](#) Calculus II ^{1, 2, 3, 4} _{_ _ _}

or [MATH 1325](#) Applied Calculus I ^{1, 2, 3, 4} _{_ _ _}

and [STAT 2332](#) Introductory Statistics for Life Sciences ^{2, 3} _{_}

[PHYS 2325](#) Mechanics ^{1, 2} _{_}

and [PHYS 2125](#) Physics Laboratory I ^{1, 2} _{_}

or [PHYS 2421](#) Honors Physics I - Mechanics and Heat ⁶ _{_}

or [PHYS 1301](#) College Physics I

and [PHYS 1101](#) College Physics Laboratory I

[PHYS 2326](#) Electromagnetism and Waves

and [PHYS 2126](#) Physics Laboratory II

or [PHYS 2422](#) Honors Physics II - Electromagnetism and Waves

and [PHYS 2126](#) Physics Laboratory II

or [PHYS 1302](#) College Physics II

and [PHYS 1102](#) College Physics Laboratory II

Major Core Courses: 26-29 semester credit hours beyond Core Curriculum

[BIOL 2281](#) Introductory Biology Laboratory ⁵ _{_}

[BIOL 2111](#) Introduction to Modern Biology Workshop I ⁵ _{_}

[BIOL 2112](#) Introduction to Modern Biology Workshop II ⁵ _{_}

[BIOL 2311](#) Introduction to Modern Biology I ^{2, 5} _{_}

[BIOL 2312](#) Introduction to Modern Biology II ⁵ _{_}

[BIOL 3401](#) Genetics

[BIOL 3402](#) Molecular and Cell Biology

[BIOL 3461](#) Biochemistry I

or [CHEM 3461](#) Biochemistry I

[BIOL 3462](#) Biochemistry II

or [CHEM 3462](#) Biochemistry II

or [BIOL 3335](#) Microbial Physiology

[BIOL 3380](#) Biochemistry Laboratory

[BIOL 4380](#) Cell and Molecular Biology Laboratory

Major Related Courses: 12 semester credit hours⁷

12 semester credit hours of upper-division [BIOL](#) electives

III. Elective Requirements: 16-19 semester credit hours

The plan must include sufficient upper-division courses to total 45 upper-division semester credit hours.

UTeach Option

The [UTeach option](#) may be added to either the BA or BS degree. UTeach Dallas Option degree plans are streamlined to allow students to complete both a rigorous Bachelor of Science or Bachelor of Arts degree and all coursework for middle or high school teacher certification in four years. Teaching Option degrees require deep content knowledge combined with courses grounded in the latest research on math and science education. While most graduates go on to classroom teaching, UTeach alums are also prepared to enter graduate school and to work in a discipline related industry.

Fast Track Baccalaureate/Master's Degrees

For students with strong academic record who intend to pursue master's studies at UT Dallas, accelerated BS/MS Fast Tracks are available. After Fast Track admission to an MS program, students may take up to 15 semester credit hours of approved graduate courses in their senior year to use toward completion of both the BS and MS degrees. The available Fast Track options and their admission requirements, if any, in addition to having completed 90 or more semester credit hours (out of which 36 hours are from the core curriculum) with a cumulative GPA of at least 3.00, are as follows:

- [MS in Molecular and Cell Biology](#): Students must have completed at least 15 semester credit hours of upper-division Biology core courses.

- [MS in Biotechnology](#): Students must have a cumulative GPA of at least 3.00 in all courses.
- [MS in Bioinformatics and Computational Biology](#): Students must have a cumulative GPA of at least 3.20 in all mathematics and statistics courses.
- [MS in Data Science and Statistics](#) (Only Data Science and Applied Statistics specializations): Students must have a cumulative GPA of at least 3.20 in all mathematics and statistics courses.

Interested students, after reviewing the [UT Dallas Fast Track policy](#), should contact their undergraduate advisor and the graduate advisor of the intended MS program well in advance of their junior year to prepare a course sequence permitting maximal advantage and apply to the Fast Track program.

Degree Planning

Upper-division biology courses taken at other institutions may be included as part of the degree plan subject to the provisions of the section on Transfer Admissions.

Major-related courses may not include more than 9 semester credit hours (BS) or 6 semester credit hours (BA) of upper-division transfer credit and not more than 3 semester credit hours (Biology major) or 6 semester credit hours (Molecular Biology major) of individual instruction (e.g., [BIOL 3V90](#), [BIOL 3V91](#), [BIOL 3V96](#), [BIOL 4302](#), [BIOL 4390](#), [BIOL 4391](#), [BIOL 4399](#), or [BIOL 4V99](#)).

Students planning a career in a particular allied health profession should consult the school they expect to attend to apprise themselves of the course requirements for admission.

Admission standards for medical and dental schools are set by the individual professional school, whose specific requirements should be reviewed with the help of the UT Dallas Health Professions Advising Center (HPAC). Most professional schools prefer that admission applications be channeled through the HPAC.

Minors

Students must take a minimum of 18 semester credit hours for the minor, 12 of which must be upper-division semester credit hours. Students who take a minor will be expected to meet the normal prerequisites in courses making up the minor, and should maintain a minimum GPA of 2.000 on a 4.00 scale (C average). Semester credit hours may not be used to satisfy both the major and minor requirements; however, free elective semester credit hours or major preparatory classes may be used to satisfy the minor.

For all minors in the School of Natural Sciences and Mathematics students must complete all prerequisite sequences for required minor courses.

Minor in Biology

18 semester credit hours

Required: 12 semester credit hours

[BIOL 2311](#) Introduction to Modern Biology I

[BIOL 2111](#) Introduction to Modern Biology Workshop I

[BIOL 3401](#) Genetics

[BIOL 3461](#) Biochemistry I

or [CHEM 3461](#) Biochemistry I

Also:

Two BIOL approved electives for majors

Minor in Biomolecular Structure

18 semester credit hours

Required: 13 semester credit hours

[BIOL 3336](#) Protein and Nucleic Acid Structure

or [BIOL 4305](#) Molecular Evolution

[BIOL 4461](#) Biophysical Chemistry (unless taken to fulfill the Molecular Biology major requirements)

[CHEM 2323](#) Introductory Organic Chemistry I

[CHEM 2325](#) Introductory Organic Chemistry II

Also:

A minimum of 5 additional semester credit hours in approved BIOL, CHEM, CS, EE, MATH, or PHYS electives

Minor in Microbiology

18 semester credit hours

Required: 15 semester credit hours

[BIOL 3520](#) General Microbiology with Lab⁸

or [BIOL 3303](#) Introduction to Microbiology⁸

and [BIOL 3203](#) Introduction to Microbiology Lab

[BIOL 3335](#) Microbial Physiology⁹

[BIOL 4350](#) Medical Microbiology

[BIOL 4345](#) Immunobiology

[CHEM 2323](#) Introductory Organic Chemistry I

Also:

One approved microbiology elective

Minor in Molecular and Cell Biology

18 semester credit hours

Required: 6 semester credit hours

[CHEM 2323](#) Introductory Organic Chemistry I

[CHEM 2325](#) Introductory Organic Chemistry II

Also:

Four approved molecular and cell biology electives

Minor in Neurobiology

18 semester credit hours

Required: 15 semester credit hours

[CHEM 2323](#) Introductory Organic Chemistry I

[CHEM 2325](#) Introductory Organic Chemistry II

[BIOL 4356](#) Molecular Neuropathology

or [BIOL 4357](#) Molecular Neuropathology II

[NSC 4353](#) Neuroscience Laboratory Methods

[NSC 4354](#) Integrative Neuroscience

or [NSC 4352](#) Cellular Neuroscience

Also:

One BIOL approved electives for majors, must be BIOL (not NSC or CHEM)

1. Curriculum Requirements can be fulfilled by other approved courses from institutions of higher education. The courses listed are recommended as the most efficient way to satisfy both Core Curriculum and Major Requirements at UT Dallas.

2. A required Major course that also fulfills a Core Curriculum requirement. Semester credit hours are counted in Core Curriculum.

3. Six semester credit hours of Calculus are counted under Mathematics Core and Component Area Option and 2 semester credit hours of Calculus are counted as Major Preparatory Courses.

4. Students may substitute MATH 2413 and MATH 2414 by taking MATH 2417 and MATH 2419.

5. Indicates a prerequisite class to be completed before enrolling for upper-division classes.

6. Students who complete PHYS 2421 do not need to complete PHYS 2125

7. Up to 3 semester credit hours of individual instruction may be used in fulfilling this requirement for BA degree. Up to 6 semester credit hours of individual instruction may be used in fulfilling this requirement for BS degree.

8. Two semester credit hours of BIOL 3520 or BIOL3303 may be used to satisfy the upper-division elective requirement for Biology and Molecular Biology majors.

9. May be substituted with CHEM 2325 Introductory Organic Chemistry II if used to satisfy the Biochemistry II core requirement for Biology and Molecular Biology majors.

Updated: 2024-08-20 09:01:40 v17.2e5c8a